

BACKGROUND

- The Komfo Anokye Teaching Hospital is one of Ghana's most premier hospitals located in the Kumasi region. It services a total population of 5,432,845 people and houses 1,200 beds
- Venipuncture is one of the most common medical procedures with 450 million annual episodes globally
- The average patient will experience 1.6-2.2 blood collections per day
- While there are vein finder devices currently on the market (AccuVein, Veinlite, Venoscope, etc.), these devices are expensive and not effective on darker skin

DESIGN CRITERIA

- We've identified five specific design criteria:
 - Cost:** device must cost less than \$300
 - Safety:** device must be usable in a manner that prevents and minimizes potential harm to user
 - Usability:** device must be functional without the need of the user to hold it
 - Functionality:** device must be effective for all (including varying BMI, ages, skin color, and skin temperatures) and must be able to operate for at least three hours before needing to be recharged
 - Durability:** device must last through consistent, everyday use as well as sterilization processes without failure

PROBLEM STATEMENT

The Komfo Anokye Teaching Hospital in Ghana takes in a large number of patients, many of which are extreme cases. Due to understaffing issues, as well as time constraints, a simple but common procedure like venipuncture can prove to be costly for hospitals and patients alike. Errors in venipuncture can cause discomfort to the patient as well as limit the efficiency of the hospital, particularly one which has limited staff and large volumes of patients.

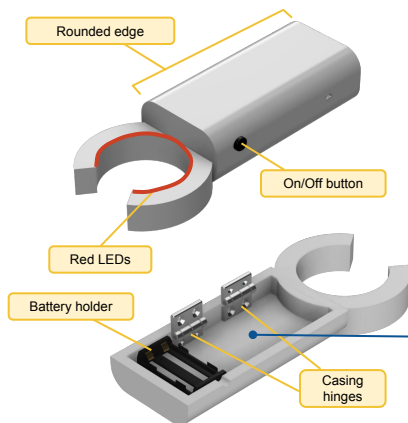
NEEDS STATEMENT

There is a need in Ghana for a **low-cost, low-energy, highly effective** vein finder device to aid physicians in venipuncture procedures.

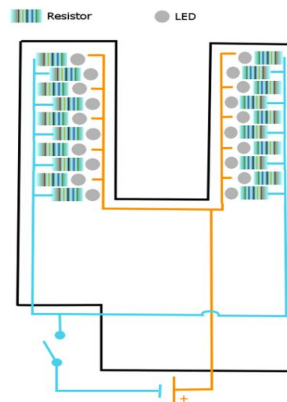
PUGH MATRIX

Criteria	Weighting	Current Device	Ideation 1: Standing NIR	Ideation 2: Strapped LED
Cost	3	0	0	1
Safety	1	0	0	0
Usability/Performance	2	0	-1	1
Durability	2	0	-1	-1
	Totals	0	-4	4

CAD DESIGN



CIRCUIT



MATERIALS AND COST

Material	Cost
Blank Circuit Board	\$5.00
Rocker Switch	\$1.40
56 Ohm Resistors	\$0.33 (x30)
LED (LED+)	\$35.00
Enclosure (3D printed, medical-grade plastic acrylic sheets)	\$12.00
Solder Wire (~1")	\$0.50
AA Batteries	\$2.00
Hinge:	\$15.00
Medical-Grade Velcro	\$9.00
LA TIM'S B-7000 Glue	\$10.00
Velcro Adjustable Strap	\$10.00
TOTAL	\$110.00

GLOBAL PARTNERSHIP

Komfo Anokye
Teaching Hospital

